

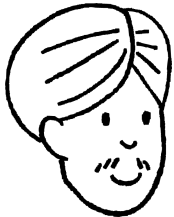


Brackets in equations

Task A: Representations of equations involving brackets

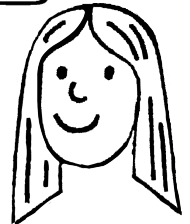
1) Write an equation and draw a bar model for Andeep and Laura's number problems.

I am thinking of a number.
When you multiply it by 5
and add 10 you get 35



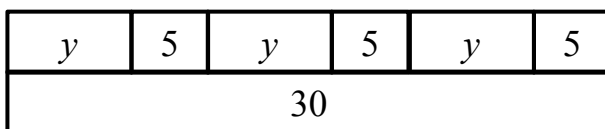
Andeep

I am thinking of a number.
When you add 10 and
multiply by 5 you get 35

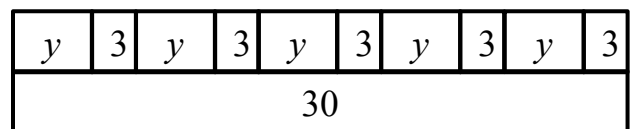


Laura

2) Write equations for the bar models that Jun and Jacob have drawn. If you can see more than one way to write the equation you should do so.



Jun



Jacob

Task B: Effectively using brackets in equations

1) Spot the errors in these manipulations of equations.

Once spotted, write a sentence to explain each error.

a) $y + 3 = 4y + 6$

$$4(y + 3) = 4y + 6$$

$$4y + 12 = 4y + 6$$

b) $y + 3 = 4y - 6$

$$2(y + 3) = 2(4y - 6)$$

$$2y + 3 = 8y - 6$$

c) $y + 3 = 4y + 6$

$$4(y + 3) = 2(4y + 6)$$

$$4y + 12 = 8y + 12$$

2) Fill in the missing parts of these manipulations of equations.

a) $2e + 3 = \frac{1}{4}e$

$$\square(2e + 3) = \square\left(\frac{1}{4}e\right)$$

$$\square + 12 = \square$$

b) $\frac{1}{2}d + 4 = 3d - 1$

$$\square\left(\frac{1}{2}d + 4\right) = \square(3d - 1)$$

$$d + \square = \square - \square$$

c) $\frac{7}{f} = 18$

$$\square\left(\frac{7}{f}\right) = \square(18)$$

$$7 = \square$$